

Coin Grabber Game

If we have a circuit with a joystick module, it should be possible to make a **simple arcade game**. While it is common to use the micro:bit LED grid as a display, it might be interesting to try using the **LCD Module** instead; we have more width (16 characters), and each character is composed of 40 pixels. Our overall resolution would be (640 x 80).

In this game, you'll **control a character with the joystick**, moving it around the LCD Module trying to **collect coins** (and possibly avoiding spikes).

Materials:



1 x Joystick Module



1 x 1602 LCD w/ i2c adapter

Wiring Diagram

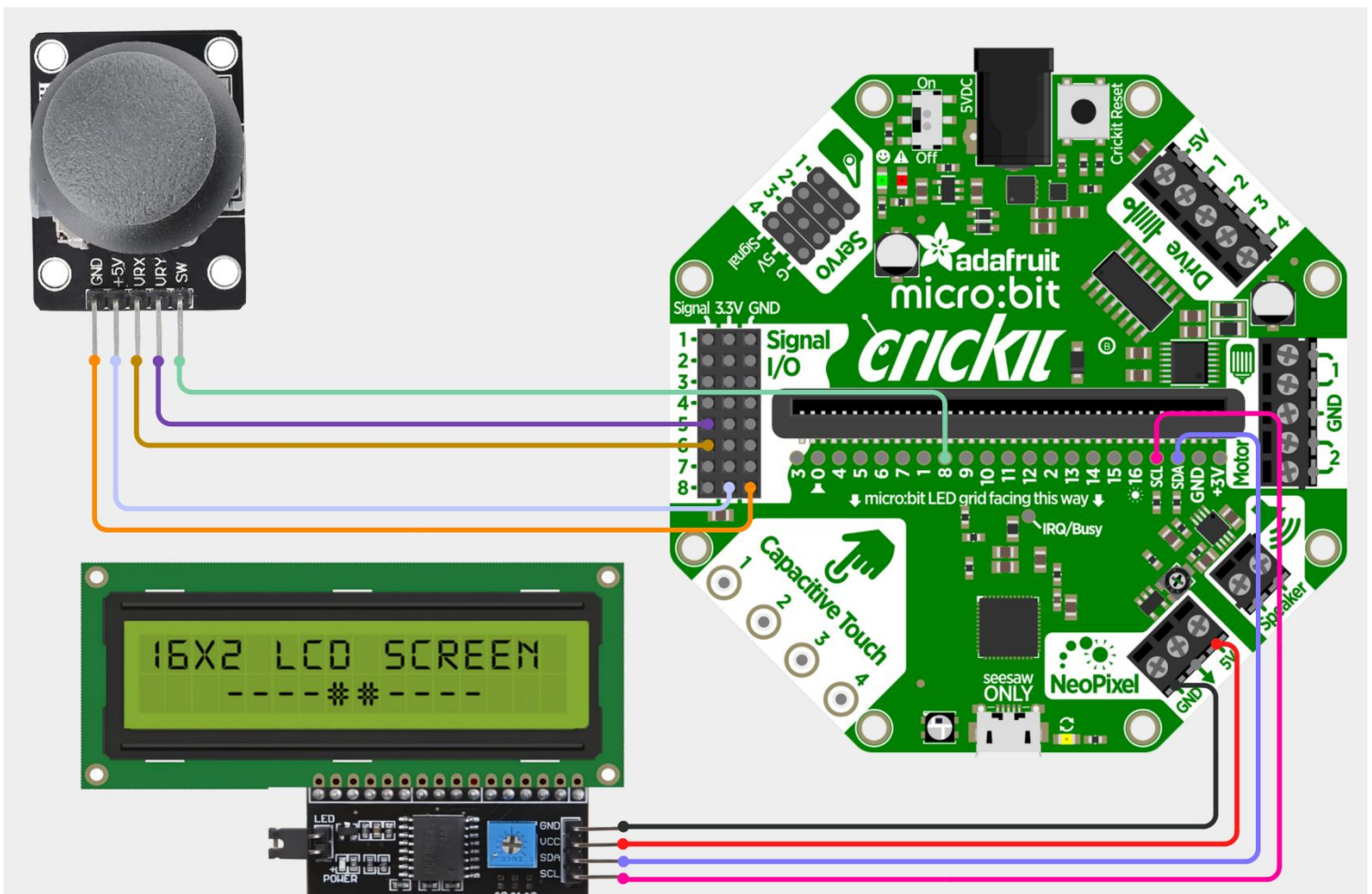
Refer to this diagram if needed to connect the joystick and lcd modules.

Signal Connections:

Joystick:

- Joystick X → Crickit Signal 6
- Joystick Y → Crickit Signal 5
- Pushbutton → Pin 8

** For this circuit enable the internal pull-up resistor on P8.*



Test One: Liquid Crystal Display

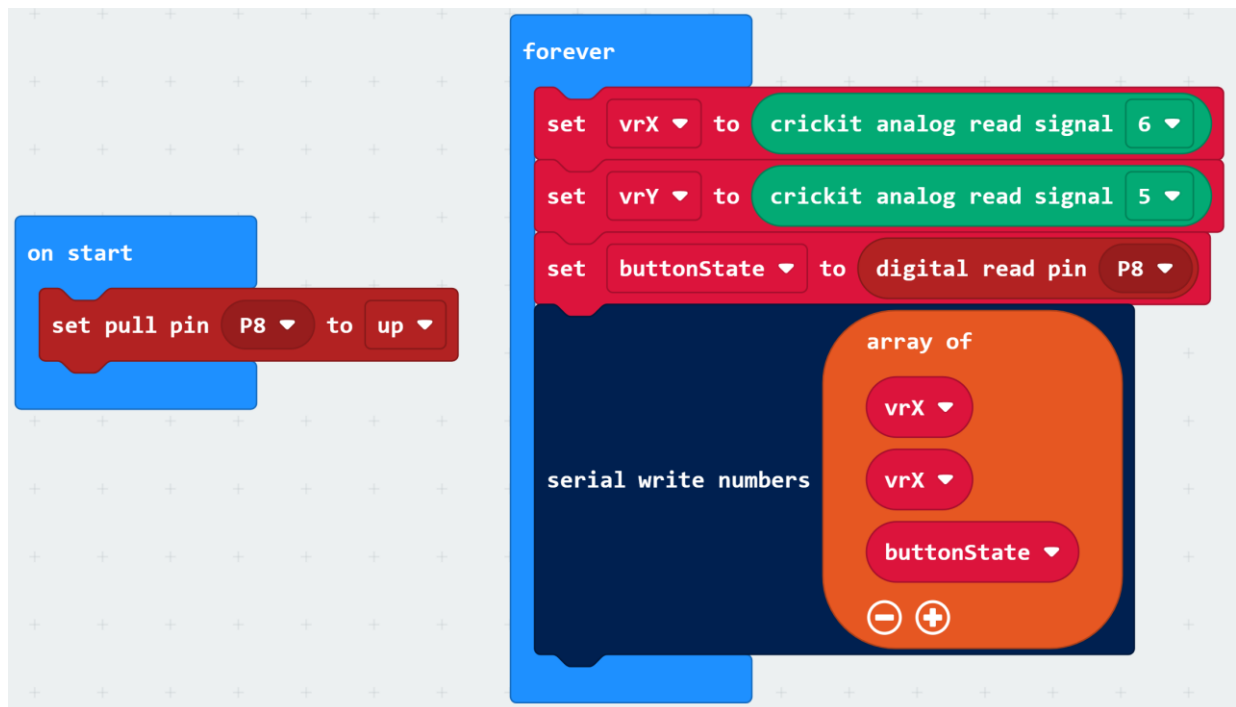
There are a several *Extensions* that allow us to communicate with the LCD. For this project, an extension that allows for **custom designed characters** would be ideal. Add the extension shown below and try the provided test blocks to ensure the component is working correctly.



Recall that you might need to adjust the potentiometer on the back of the module to improve the contrast if the text is difficult to see.

Test Two: Joystick Module

Here we want to verify the values we read in (x/y rotation, and pushbutton):

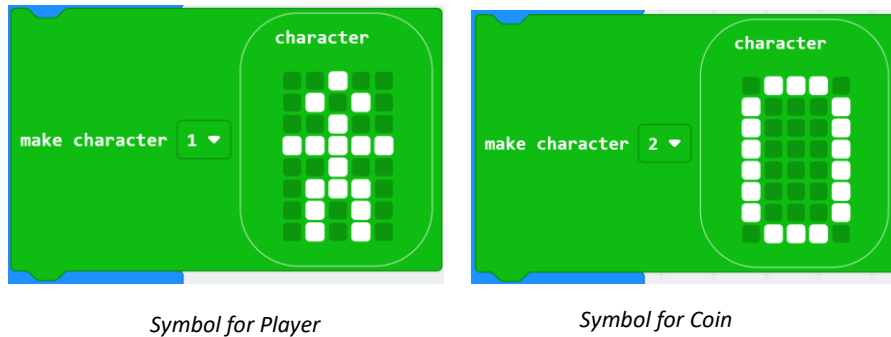


Assignment Details: Base Features

For this assignment, complete the following base features:

Create the Assets, and their Variables

Begin by designing a character and a coin as **special symbols**. One possible solution could be:



For both the player and the coin, we should have some variables to track their positions:



Initially, make sure the coins and the player are on the same row ($y == 1$, for instance). Later on jumping can be added...

Create the Game Loop

Try to encapsulate the logic for the game in a **single forever loop** if possible. The following pseudocode can be followed as a guide:

ON START:

- Set up initial values for position variables (player and coin).
- Draw the **special characters** (for player and coin) on the LCD module.

FOREVER:

- Check for horizontal movement with joystick module
 - o If LEFT or RIGHT movement happens:
 - Erase the current player tile
 - Update **playerX**
 - Re-draw the player at new playerX, player
- Check if the coin and player are located at the same spot

- `coinX == playerX` **and** `coinY == player`
 - if that is true, generate new `coinX`, `coinY` values and redraw the coin at its new location.
 - `score += 1`
- Insert Pause (somewhere between 100ms and 500ms). *This will control the max movement speed.*

Some Nice Extras

- Add the ability to jump.
 - When the joystick button is pressed, set the **playerY** = 0
 - After a set amount of “jump time” set **playerY** = 1;
 - Make sure to re-draw the player at the new coordinate each time player is updated.
- Add spiky obstacles.
 - On the bottom level, add a randomly placed spike (custom character) that you must avoid.

----- Suggestions and Hints are below but try to build what you can on your own first! -----

Hints and Suggestions:

